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Preface

About Version 1.1

It has been 8 years since this collection was put together and published as *Introduction to Physics* through the OpenStax CNX platform (previously Connexions). The textbook was continuously edited, reaching version 12.1, from initial publication until the 2020 announcement of the planned retirement of the CNX platform. That version is archived on GitHub by OpenStax. The collection has undergone hosting changes (there is a version on LibreTexts with shifted chapter and section numbers), and it is now self-hosted at the canonical URL intro.coaphys.xyz with a GitHub repository (concepts-of-physics-2e) for version and release control. This Version 1.1 is the first major revision of this OER collection in 6 years.

Some of the changes made for Version 1.1 are a continuation of the work that was interrupted by the retirement of the CNX platform, as I was in the process of curating the chapter-end exercises that were better suited for this conceptual physics collection. The partial completion of this curation can be seen in earlier versions preserved elsewhere, with some chapters already containing mostly conceptual questions and other chapters containing an unmodified collection of questions from OpenStax *College Physics*. Other changes are made to reflect the updated data, new discoveries and missions, and major developments such as the 2019 redefinition of SI base units, particularly the kilogram and the ampere. The most substantial revisions, in Unit 4 and especially in the Special Relativity chapter (which previously used the *College Physics* modules unmodified), correct major conceptual mistakes in the earlier version (which had been reported to OpenStax earlier).

Recent developments in technology both made possible and facilitated the improvements you will see in this edition. First, this self-hosted edition would not be possible without the OpenAI Codex tool, as I am primarily trained as a physicist, not a web developer. If this work were being done before the advent of modern AI tools, this edition would have been put together as a collection of Microsoft Word files. Second, numerous technical fixes (especially correcting MathML expressions, which I was editing by hand in 2018) were now done essentially autonomously with Codex, using model GPT-6 Astra. Lastly, most of the textbook prose revisions (both for correcting conceptual errors and for clearer presentation) were done using GPT-6 Astra, always reviewed by me, and some of the more challenging sections still directly drafted by me. As illustrative examples, in percentage terms, these five sections were most heavily revised, and a reliable checker for AI text such as Pangram will show substantial portion of AI-written prose: Einstein's Postulate, Biological Effects of Ionizing Radiation, Uncertainty Principle, Statistical Interpretation of Entropy and the Second Law of Thermodynamics: The Underlying Explanation, and GUTs: The Unification of Forces.

I hope this version will continue to be useful to instructors and students looking for an openly licensed (this collection continues to be licensed under CC BY 4.0) textbook for a college-level general-education conceptual physics course. One note on the chapter and section numbering: the web version of this edition will allow you to toggle between "Lecture-aligned" numbering (a.k.a. LibreTexts numbering) and "CNX-aligned" numbering. While the long-term plans are in place to switch back to "CNX-aligned" numbering (it's the correct numbering scheme for the chapters and sections), the PDF version of this edition will use "Lecture-aligned" numbering until the recorded lectures at College of Alameda can be updated to use the "CNX-aligned" numbering. Much care was taken to make internal inter-section references independent of the numbering system.

Andrew Park, Physics Instructor at College of Alameda, Fall 2026

Original “About This Book”

Introduction to Physics is an adaptation of *Concepts of Physics*, itself an adaptation based on modules from OpenStax *College Physics*. Some of the modules removed from *College Physics* have been put back in, most notably modules on momentum, rotation and special relativity. Sections on linear momentum has been added from James Rittenbach’s derived copy of *Concepts of Physics*. Additional minor edits have been made as necessary.

This textbook is intended for a one-semester introduction to physics course, requiring as little mathematics as possible. Students are expected to have taken some algebra and geometry at high-school level, but no higher level of mathematics is required (and use of algebra and geometry is minimized as much as possible, in order to emphasize the concepts of physics). The courses this textbook is intended for often go by the name of “conceptual physics”, “descriptive physics”, or “introduction to physics”.

This edition is put together for use by College of Alameda’s Physics 10 course, starting in Summer 2018 (updated and revised as needed thereafter). Much gratitude is due to Prof. Bobby Bailey for putting together *Concepts of Physics* and to OpenStax for making this Open Educational Resource available.

Andrew Park, Physics Instructor at College of Alameda, June 2018

P.S. Original “Preface” to *Concepts of Physics* follows below.

Preface to Concepts of Physics

Concepts of Physics is an adaptation based on modules from OpenStax College’s textbook *College Physics*.

This text is intended for use in a one-semester physical science course that requires algebra but no trigonometry.

Not all physics topics are covered in this work. Of the nearly 300 original *College Physics* modules, only 100 were retained. For these selected modules, content was modified (as needed) to remove/replace any trigonometry-related topics, examples, homework, or figures. Occasionally, original content was edited (or replaced) to reflect the level, and coverage, of the course. Notation was modified in a few cases for consistency between modules. Explicit links between modules (and discussions related to non-retained modules) were removed. Finally, all direct links to PHeT plugin-based (Java or Flash) physics simulations were removed.

The authors of *College Physics* are commended for providing such a comprehensive body of high-quality material. Special thanks to them and to OpenStax. Additional thanks to University of Maryland University College (UMUC) and Debra F. McLaughlin.

Bobby Bailey, January 2015

About OpenStax College

OpenStax College is a non-profit organization committed to improving student access to quality learning materials. Our free textbooks are developed and peer-reviewed by educators to ensure they are readable, accurate, and meet the scope and sequence requirements of modern college courses. Unlike traditional textbooks, OpenStax College resources live online and are owned by the community of educators using them. Through our partnerships with companies and foundations committed to reducing costs for students, OpenStax College is working to improve access to higher education for all. OpenStax College is an initiative of Rice University and is made possible through the generous support of several philanthropic foundations.

Sources and adaptations

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This source was based on: Preface to Concepts of Physics, by Bobby Bailey.

This section was recovered from *Introduction to Physics*, CNX collection col25183, version 12.1, and subsequently revised. See the source repository and revision history.